MATH 152 **QUIZ 10A** NAME Solutions

Mrs. Bonny Tighe 25 points

10.3 TA_____ Wed 5/8/2024

$x = r\cos\theta \quad y = r\sin\theta$

1. Find the equation of the tangent line to the curve $r = 2 - \cos 2\theta$ when $\theta = \pi/6$

$r(\pi/6) = 2 - \cos \pi/3 = 2 - 1/2 = 3/2$ $\quad x = 3/2 \cos \pi/6 = 3/2(\sqrt{3}/2)$

$r'(\theta) = 2\sin 2\theta \quad r'(\pi/6) = 2\sin \pi/3 = \sqrt{3}$ $\quad y = 3/2 \sin \pi/6 = 3/2(1/2)$

$$\frac{dy}{dx} = \frac{r\cos\theta + \sin\theta\, r'(\theta)}{r(-\sin\theta) + \cos\theta\, r'(\theta)} = \frac{3/2(\sqrt{3}/2) + 1/2(\sqrt{3})}{3/2(-1/2) + \sqrt{3}/2(\sqrt{3})} = \left(\frac{4}{4}\right)$$

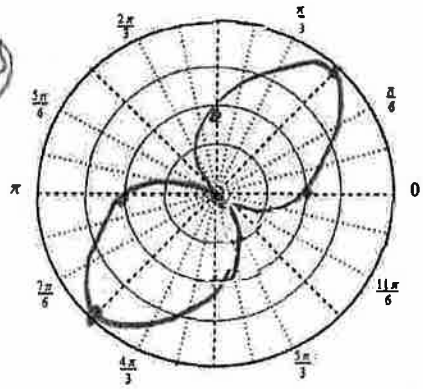
$$m = \frac{3\sqrt{3} + 2\sqrt{3}}{-3 + 2(3)} = \frac{5\sqrt{3}}{3} \qquad (\sqrt{3}/4,\ 3/4)$$

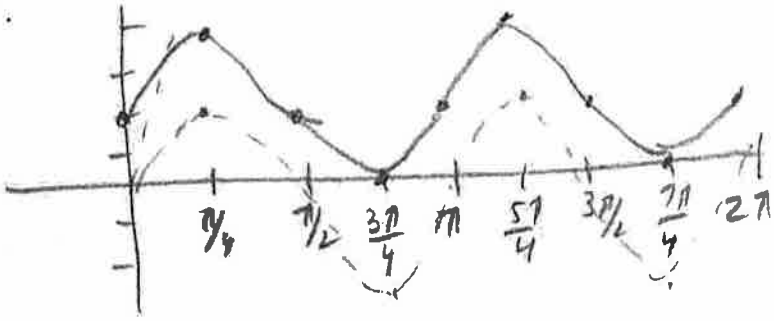
7 pts

$$y - 3/4 = \frac{5\sqrt{3}}{3}(x - \sqrt{3}/4)$$

2. Sketch the graph. $r = 2 + 2\sin 2\theta$.

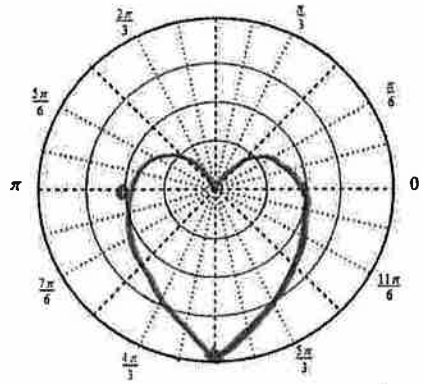
4 pts

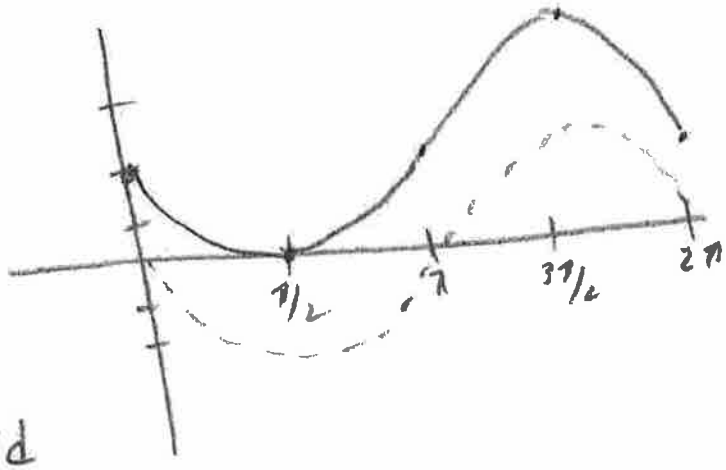




3. Sketch the graph $r = 2 - 2\sin\theta$

4pts

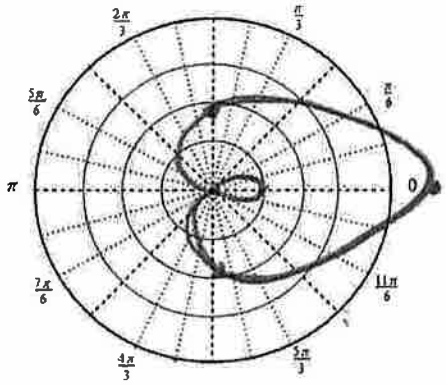


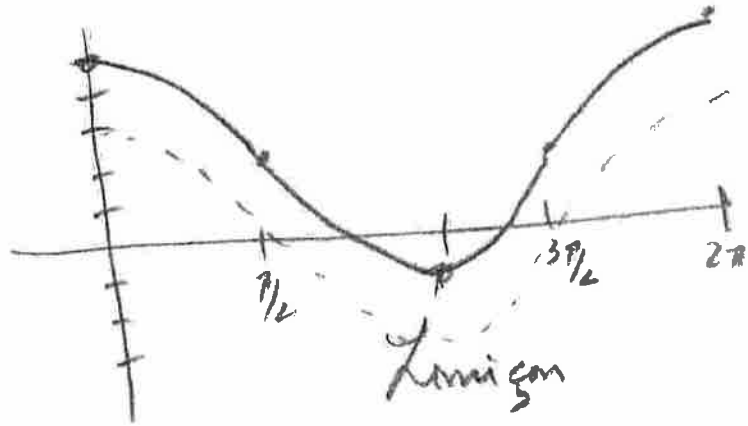


Cardioid

4. Sketch the graph $r = 2 + 3\cos\theta$

4pts





Limaçon

5. a) Find the Cartesian coordinates for the polar coordinate $(-3, 7\pi/4)$ $\left(-3\sqrt{2}/2,\ 3\sqrt{2}/2\right)$

2pts $x = -3\cos 7\pi/4 = -3\sqrt{2}/2$ $\quad y = -3\sin 7\pi/4 = -3(-\sqrt{2}/2) = 3\sqrt{2}/2$

b) Find polar coordinates for the Cartesian coordinate $(-2, 2\sqrt{3})$ where $r < 0$, *and* $\theta < 0$ $\left(-4,\ -\pi/4\right)$

$\tan\theta = \frac{2\sqrt{3}}{-2} = -\sqrt{3}$

2pts $r^2 = (-2)^2 + (2\sqrt{3})^2$

$= 4 + 4(3) = 16$ $\quad r = 4$

c) Give two alternate versions on the polar point $(-2, -5\pi/4)$

2pts

$(-2, 3\pi/4)$ $\quad (2, -\pi/4)$

$(2, 7\pi/4)$ etc.

MATH 152 **QUIZ 10** NAME Solutions

Mrs. Bonny Tighe 25 points

10.3 TA_____ ~~Wd~~ 5/8/2024

1. Find the equation of the tangent line to the curve $r = 1 - \sin 2\theta$ when $\theta = \pi/3$

$x = r\cos\theta$

$x = (1-\sqrt{3}/2)(1/2) = (1/2 - \sqrt{3}/4)$

$y = r\sin\theta$

$y = (1-\sqrt{3}/2)(\sqrt{3}/2) = (\frac{\sqrt{3}}{2} - \frac{3}{4})$

7 pts

$r(\pi/3) = 1 - \sin(2\pi/3)$

$1 - \sqrt{3}/2$

$r'(\theta) = -2\cos 2\theta$

$r'(\pi/3) = -2\cos(2\pi/3)$

$= -2(-1/2) = 1$

$$dy/dx = \frac{r\cos\theta + \sin\theta\, r'(\theta)}{r(-\sin\theta) + \cos\theta\, r'(\theta)}$$

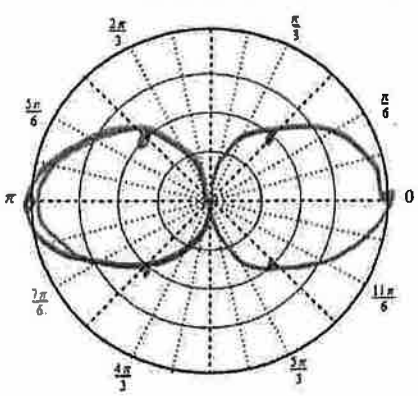
$$m = \frac{(1-\sqrt{3}/2)(1/2) + \sqrt{3}/2(1)}{(1-\sqrt{3}/2)(-\sqrt{3}/2) + (1/2)(1)} = \frac{\frac{1}{2} - \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2}}{-\frac{\sqrt{3}}{2} + \frac{3}{4} + 1/2}\left(\frac{4}{4}\right)$$

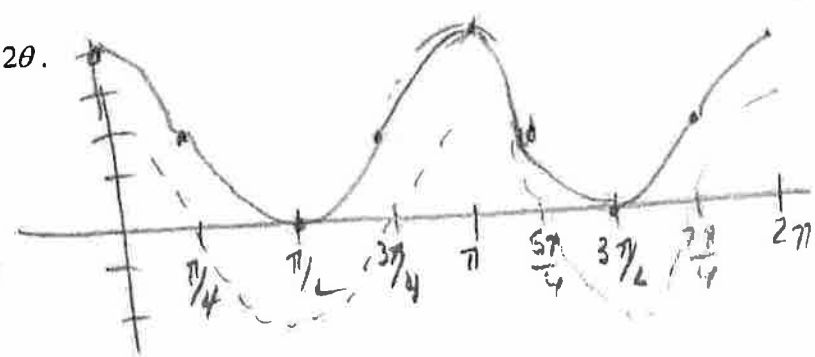
$$m = \frac{2 - \sqrt{3} + 2\sqrt{3}}{-2\sqrt{3} + 3 + 2} = \frac{2+\sqrt{3}}{5-2\sqrt{3}}$$

$$y - (\sqrt{3}/2 - 3/4) = \frac{2+\sqrt{3}}{5-2\sqrt{3}}\left(x - (\frac{\sqrt{3}}{2} - 3/4)\right)$$

2. Sketch the graph. $r = 2 + 2\cos 2\theta$.

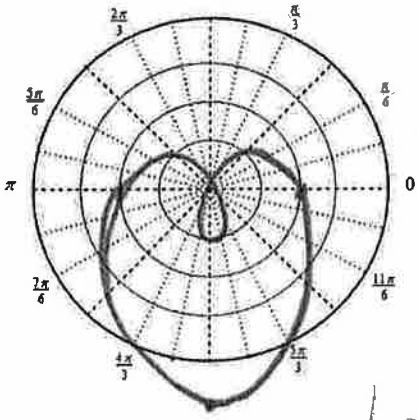
4 pts

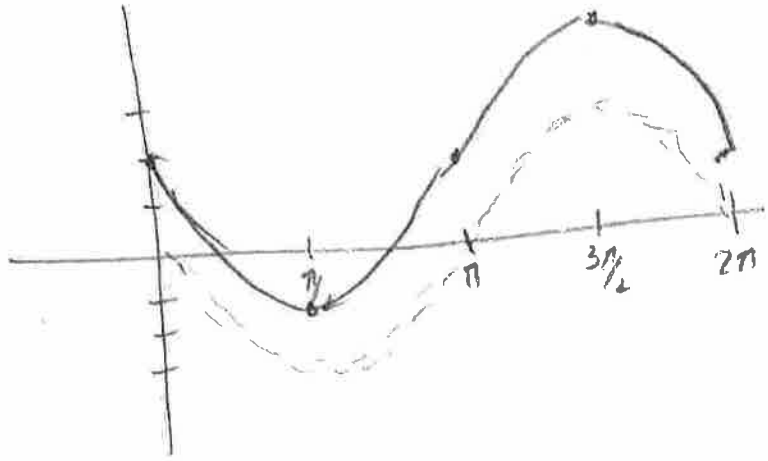




3. Sketch the graph $r = 2 - 3\sin\theta$

4pts

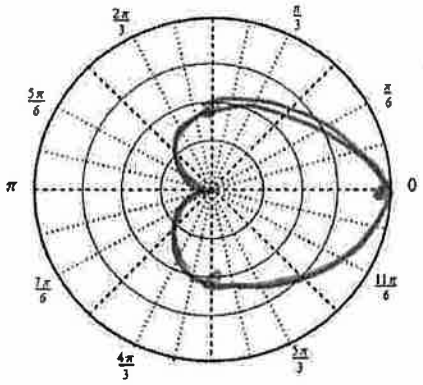


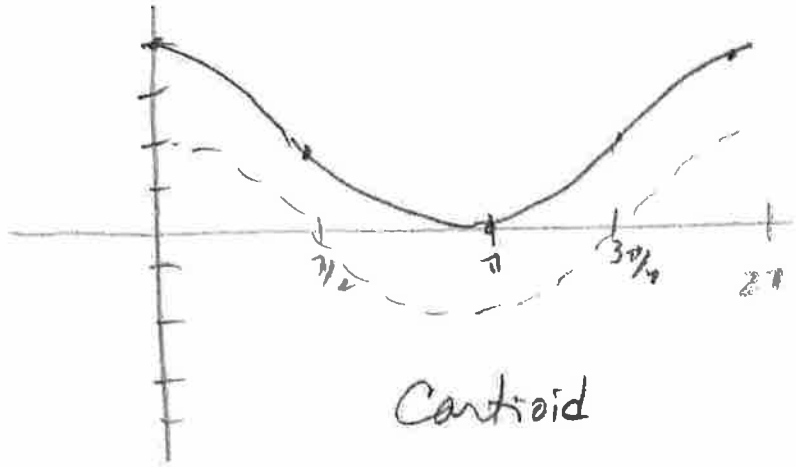


Limaçon

4. Sketch the graph $r = 2 + 2\cos\theta$

4pts





Cardioid

2pts

5. a) Find the Cartesian coordinates for the polar coordinate $(-2, 5\pi/6)$

$(\sqrt{3}, -1)$

$x = -2\cos 5\pi/6 = -2(-\sqrt{3}/2) = \sqrt{3}$

$y = -2\sin 5\pi/6 = -2(1/2) = -1$

2pts

b) Find polar coordinates for the Cartesian coordinate (-2, 2) where $r < 0, and\ \theta < 0$

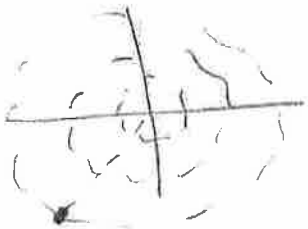
$r^2 = (-2)^2 + (2)^2 = 4 + 4 = 8$

$r = 2\sqrt{2}$

$(-2\sqrt{2}, -\pi/4)$

c) Give two alternate versions on the polar point $(3, 5\pi/4)$

2pts



$(-3, \pi/4)$

$(3, -3\pi/4)$

$(-3, -7\pi/4)$

etc.